LESSON PLAN

AIRCRAFT PROPULSION & ENGINEERING SYSTEM (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	12
Total No. of Lesson Plans	14	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Methods of applying surface protection, preparation of paints, equipment used, common paint and troubles
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Methods of applying surface protection, preparation of paints, equipment used, common paint and troubles.
5. Specific Objectives	6.2.1.1 Trainee will be able to understand surface protection materials, paint identification, restoration of various paint and application, dope shop, dope articles and troubles.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of paint preparation and application techniques using standard equipment; Safety precautions include PPE usage and ventilation; Instructor supervision required for all mixing and spraying activities.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary functions of surface protection materials?
(b) List three common types of paints used in aircraft maintenance.
(c) Why is surface preparation critical before applying paint?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Surface protection materials overview • Paint identification and coding • Restoration of existing paint systems • Standard application methods • Dope shop operations and articles	L & Ds	PPT, IB	15 min
(b)	• Paint preparation procedures • Equipment usage and maintenance • Common paint troubles and defects • Troubleshooting application issues • Safety protocols during application	L & Ds	PPT, PDF	20 min
(c)	• Review of paint identification codes • Analysis of common paint defects • Corrective actions for application errors • Proper storage of paint materials • Final inspection criteria	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the correct procedure for preparing a paint mixture for application.
(b) Identify two common paint defects and their likely causes.
(c) Describe the key safety measures required when operating paint equipment.

12. Home Assignment and Follow Up	Prepare a brief report on the differences between dope articles and standard paints, referencing the next lesson plan on colors and marking on naval aircraft.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Colors and marking on naval aircraft.	Refer to the master syllabus reference list	LP 13

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min